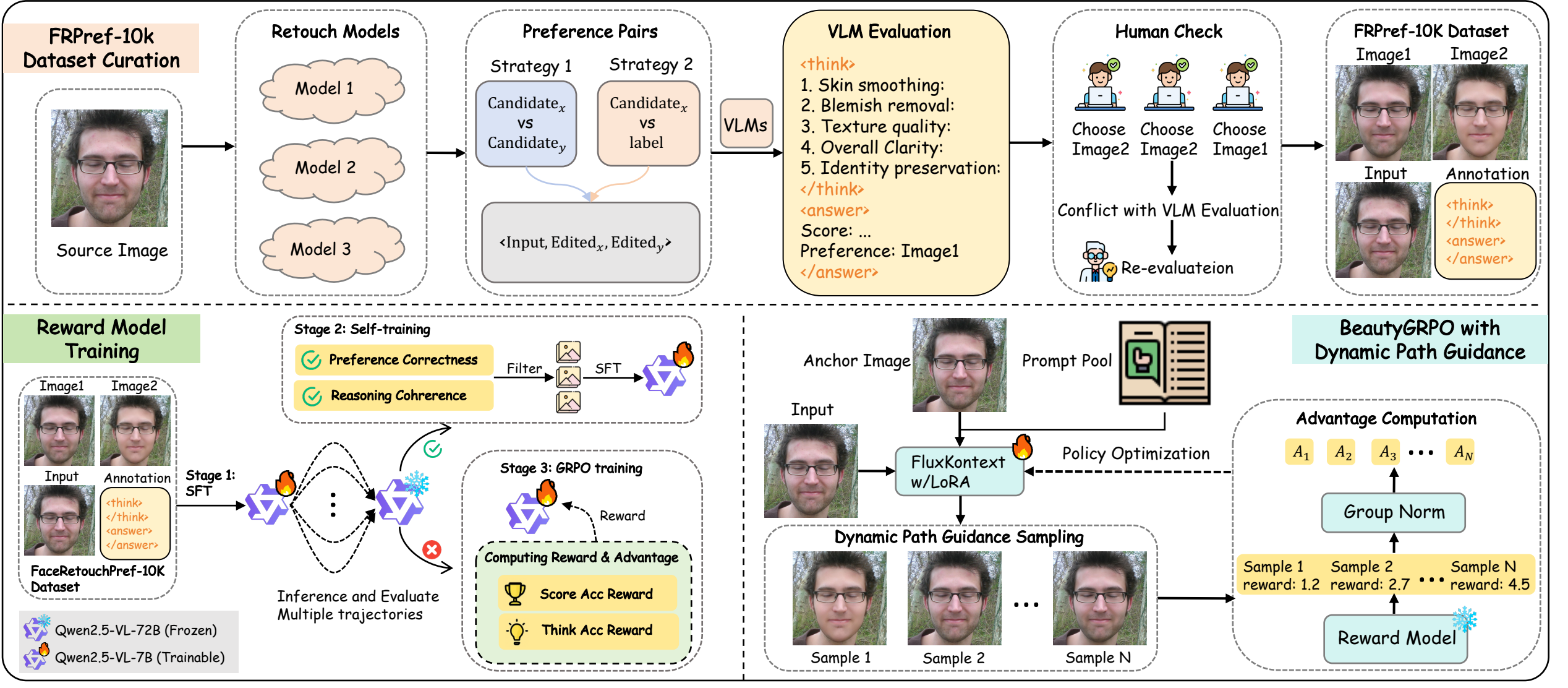
FRPref-10k Dataset Curation
Source Image
Retouch Models
Model 1
Model 2
Model 3
Preference Pairs
Strategy 1
Strategy 2
Candidate$_x$ vs Candidate$_y$
Candidate$_x$ vs label
<Input, Edited$_x$, Edited$_y$>
VLMs
VLM Evaluation
<think>
1. Skin smoothing:
2. Blemish removal:
3. Texture quality:
4. Overall Clarity:
5. Identity preservation:
</think>
<answer>
Score: ...
Preference: Image1
</answer>
Human Check
Choose Image2
Choose Image2
Choose Image1
Conflict with VLM Evaluation
Re-evaluateion
FRPref-10K Dataset
Image1
Image2
Input
Annotation
<think>
</think>
<answer>
</answer>
Reward Model Training
Image1
Image2
Input
Annotation
<think>
</think>
<answer>
</answer>
FaceRetouchPref-10K Dataset
Stage 1: SFT
Stage 2: Self-training
Preference Correctness
Reasoning Cohrerence
Filter
SFT
Inference and Evaluate Multiple trajectories
Stage 3: GRPO training
Reward
Computing Reward & Advantage
Score Acc Reward
Think Acc Reward
Qwen2.5-VL-72B (Frozen)
Qwen2.5-VL-7B (Trainable)
BeautyGRPO with Dynamic Path Guidance
Anchor Image
Prompt Pool
Input
FluxKontext w/LoRA
Policy Optimization
Dynamic Path Guidance Sampling
Sample 1
Sample 2
Sample N
Advantage Computation
A_1 A_2 A_3 ... A_N
Group Norm
Sample 1 reward: 1.2
Sample 2 reward: 2.7
Sample N reward: 4.5
Reward Model